ALICE L WALTON SCHOOL OF MEDICINE

Aortic Dissection

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Adapted from: Aortic Dissection

Listen to this Brick

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Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Differentiate the types of aortic dissection.
2. Differentiate the clinical presentation of aortic dissection.
3. Explain the pathophysiology and resulting consequences of aortic dissection.
4. Interpret cli [illegible] ic dissection.
5. Outline the management plan of aortic dissection.

CASE CONNECTION

KG, a 59-year-old male, is the fourth patient you've seen today with back pain. You think, 'The first two had musculoskeletal strain, and the other had a ruptured lumbar disc, but this patient is different. I'm not sure what's going on. He says the pain is in his upper back and feels like something is being torn apart inside of him, like a piece of cloth being ripped in half.' His exam is normal, including his musculoskeletal and neurologic exam, but his blood pressure (BP) is 168/100 mm Hg. Even his ECG and chest x-ray are normal. 'I think we need to do additional testing,' you think.

What diagnostic test(s) would you advise? What treatment would you recommend? Consider your answers as you read, and we'll revisit KG at the end of the brick.

GO TO CONCLUSION ↓

What is an Aortic Dissection? (LO #1)

Dissection is separation of a blood vessel wall's layers. It occurs almost exclusively in arteries, usually the aorta or arteries in the head. This often leads to the formation of a hematoma within the vessel wall, which can enlarge and occlude blood flow through the vessel or compress nearby structures.

Types of Dissectio [illegible]

An aortic dissection is categorized based on whether it involves the ascending aorta. If so, it is a Stanford type A dissection, regardless of how far distally the dissection travels. All other aortic dissections are Stanford type B (Figure 1).

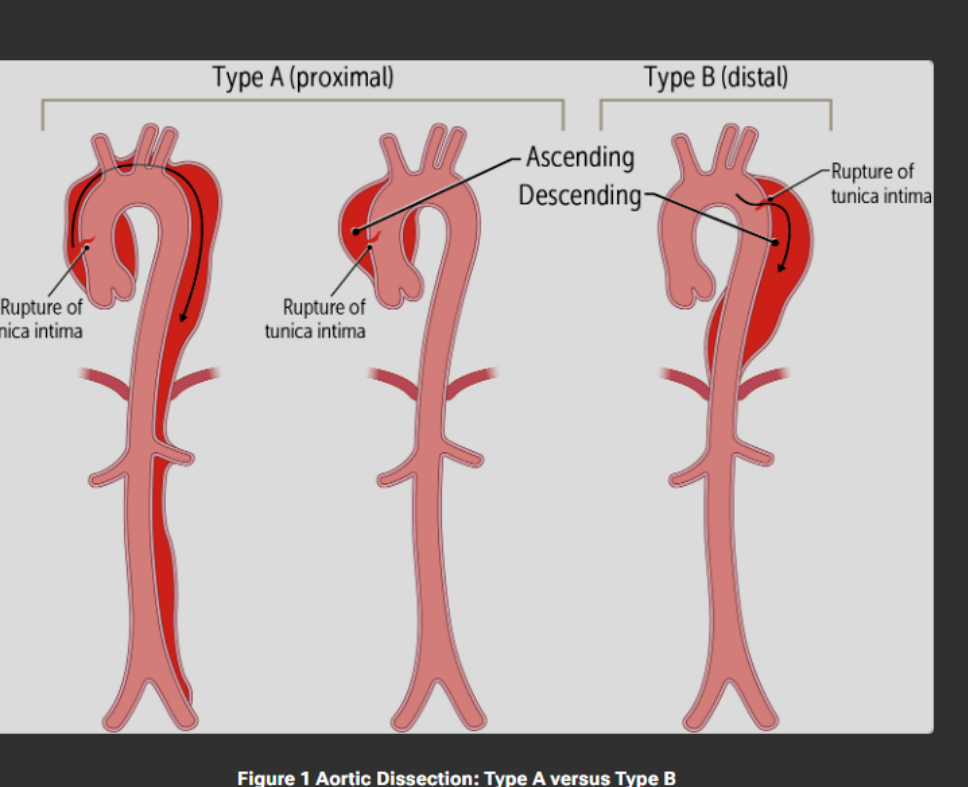


Figure 1 Aortic Dissection: Type A versus Type B

Stanford type A aortic dissections are more common and more dangerous. Besides classifying the dissections, the Stanford system separates patients who need immediate surgical intervention (type A) from those who can be managed conservatively with d [illegible]

- **Stanford type A**
  - Involves the ascending aorta with or without descending involvement
  - More common and more dangerous
  - Needs immediate surgery
- **Stanford type B**
  - Does not involve the ascending aorta (i.e. descending aortic involvement only)
  - Can be managed conservatively with drugs

Epidemiology

The incidence of aortic dissections is 2 to 3 per 100,000 patients (about half that of aortic aneurysm, the other common aortic disorder). It's most common in older males (age 60-80 years). Older patients typically have more atherosclerotic risk factors (eg, smoking, diabetes, hypertension), whereas younger patients most often have a congenital disorder of collagen (a component of the vessel wall).

What distinguishes a Stanford type A aortic dissection from a type B?

How Do Patients With Aortic Dissection Present? (LO #2)

The most common presentation of aortic dissection is **sudden-onset severe chest or back pain.** The pain is classically described as "ripping" or "tearing." Pain usually involves the chest in Stanford type A aortic dissections and the back in Stanford type B. However, a few patients present atypically, including some without any chest or back pain.

Other symptoms may be related to changes in the patient's blood pressure (BP). For example, syncope, hypotension, and shock may be seen in type A, whereas symptoms related to severe hypertension (e.g., neurologic changes or heart failure) may be seen in type B.

We'll discuss symptoms related to other complications (aortic valve damage, pericardial effusion) in the following sections.

What is the most common presentation of a patient with aortic dissection?

What Is the Pathophysiology of Aortic Dissection? (LO #3)

Aortic dissection has a variety of causes but leads to some characteristic pathology and complications.

Causes

The causes of aortic dissection vary depending on age. In the most common dissections seen in older patients, disease is caused by severe, poorly controlled **hypertension**. A smaller group of older patients may instead have weakening of the arterial wall caused by inflammation (e.g. from a vasculitis such as Takayasu arteritis). Severe chest trauma is an unusual cause.

In the less-common cases seen in younger patients, aortic dissection is usually the result of trauma or connective tissue disease, such as Marfan or Ehlers-Danlos syndrome, with defects in elastin and collagen synthesis, respectively, both of which are required for healthy aortic walls.

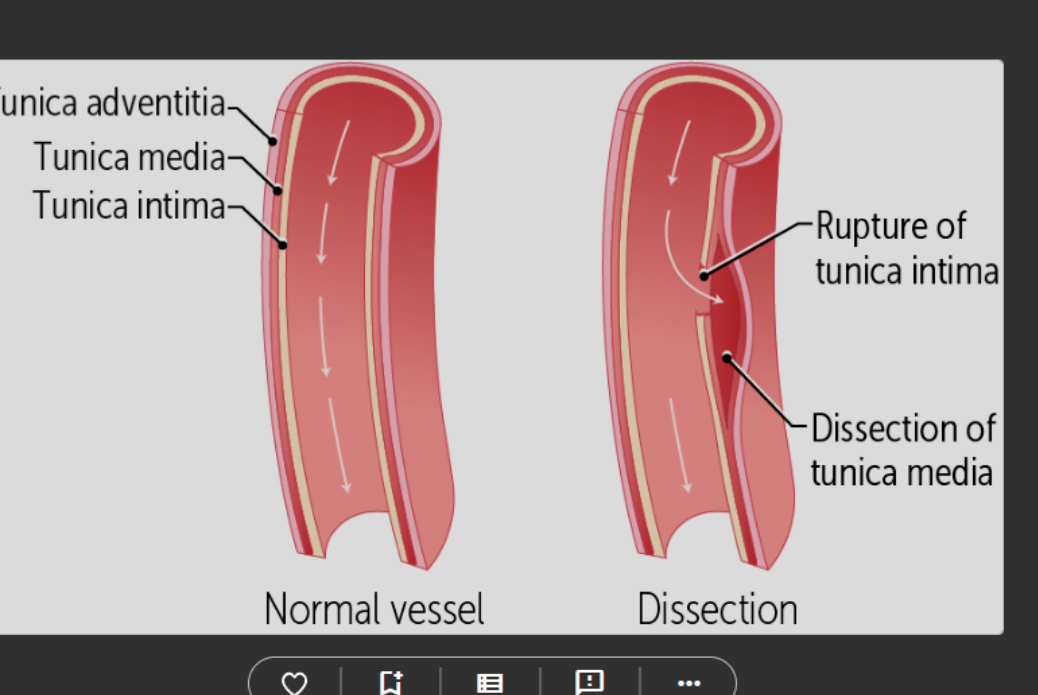
CLINICAL CORRELATION

Marfan syndrome and Ehlers-Danlos syndrome are genetic diseases characterized by abnormal and we [illegible] iseases are associated with an increased risk for aortic dissection.

Other risk factors include older age, male sex, and (in type A) having a bicuspid aortic valve.

Pathology

Regarding the etiology and resulting consequences of aortic dissection, some are common to both types of aortic dissections, and others are unique to Stanford type A. In both types, the principal cause is degeneration of the aortic wall's **tunica media** (Figure 2).



In Figure 2, the left image shows normal blood flow; the right image shows blood entering the vessel wall via a tear in the tunica intima. This tear allows blood to collect within the tunica media, where it "unzips" the layers of smooth muscle down the length of the vessel (recall KG from the Case Connection feeling like something was being ripped in two). This allows a false lumen (a "parallel road") of blood to build up within the vessel wall (specifically within the tunica media), forming a hematoma that can enlarge and occlude blood flow through the true lumen of the artery. The hematoma is contained only by the outer tunica adventitia.

Cystic medial degeneration (Figure 3) is a pathologic alteration of the vascular medial layer, commonly associated with **hypertension** or **connective tissue diseases** (e.g., Marfan syndrome). CMD is the histologic lesion most frequently detected in cases of aortic dissection, characterized by smooth muscle cell dropout and necrosis, elastic tissue fragmentation, and accumulations of amorphous proteoglycan-rich extracellular matrix. However, recognizable medial damage is not a prerequisite for dissection, as in may cases no specific underlying defect is identified in the arterial wall.

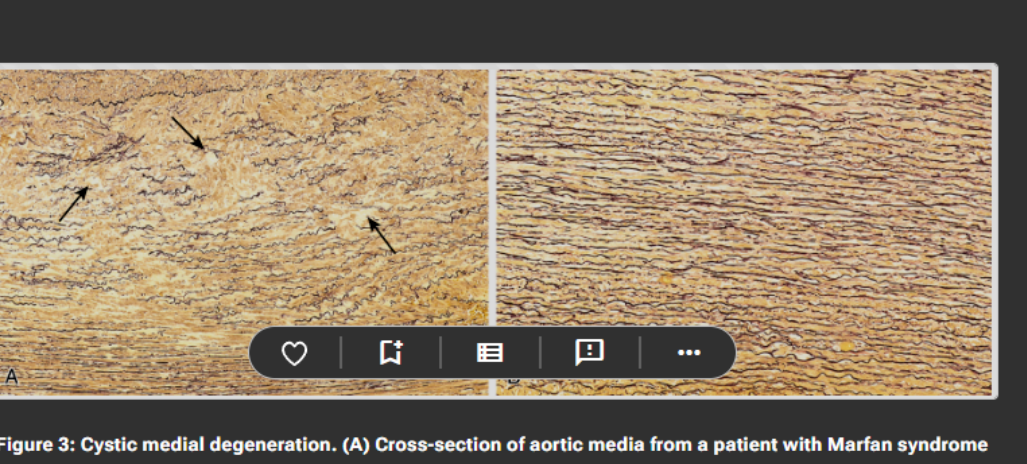


Figure 3: Cystic medial degeneration. (A) Cross-section of aortic media from a patient with Marfan syndrome showing marked elastin fragmentation and formation of areas devoid of elastin that resemble cystic spaces but are actually filled with increased amounts of proteoglycans *(arrows)*. (B) Normal media for comparison showing the regular layered pattern of elastic tissue. In both (A) and (B), elastin is stained black.

With aortic dissection, in which vascular layer does blood accumulate?

Physiologic Consequences and Complications

In addition to pain, a minority of patients may present with diminished pulses, heart murmurs, or neurologic deficits resulting from blockage of involved branches of the aorta or compression of surrounding structures by the enlarging aorta.

Blockage of Blood Flow. Dissections can block blood flow through the aorta or its branching vess [illegible] he "flap" of intima created by the dissection may extend into the branching vessel and occlude the lumen, blocking blood flow. Second, the hematoma that forms within the false lumen created by the dissection can compress the openings of the arteries that branch off the aorta, leading to end-organ ischemia. These blockages can involve the carotid, subclavian, spinal, or other visceral (eg, mesenteric) arteries.

How might these blockages be detected clinically? Patients may show a variation in pulse or BP between their two arms, caused by obstruction of the subclavian arteries. This can occur in both type A and type B dissections.

If the obstruction involves the aortic root vessels (usually type A dissections), limited cerebral blood flow may instead result, with consequences ranging from dizziness and syncope to stroke with neurologic deficits (eg, weakness, numbness, and dysarthria).

Recall that the coronary arteries emerge from the very first (proximal) part of the ascending aorta. In a type A dissection, if the coronary arteries are blocked, cardiac blood flow may be limited, and chest pain or myocardial infarction can result.

Aortic Rupture. The feared progression of aortic dissection to complete **aortic rupture** results in massive internal bleeding that is usually fatal.

Rupture can also occur into the pericardium, leading to **hemopericardium** and **cardiac tamponade**. This will cause hypotension, jugular venous distention (JVD), and distant heart sounds. This is why a Stanford type A aortic dissection is considered more dangerous than is a Stanford type B dissection; the risk of spreading to the heart makes Stanford type A dissection a true surgical emergency.

Compression of Adjacent Structures. The enlarging dissection can compress adjacent structures, such as the:

- Superior cervical (sympathetic nerve) ganglia, leading to Horner syndrome (triad of ptosis, miosis, anhidrosis)
- Recurrent laryngeal nerve, leading to hoarseness

CLINICAL CORRELATION

Compression of the sympathetic nerves that innervate the head, eye, and neck can lead to a triad of symptoms including dropping eyelid (ptosis), a constricted pupil (miosis), and lack of sweating of the face (anhidrosis). This constellation of symptoms is referred to as Horner syndrome.

Aortic Valve Disease. A type A aortic dissection can spread and extend in retrograde fashion to the aortic valve, leading to a tear in the aortic valve annulus. This may lead to acute aortic valve regurgitation, with associated heart failure and hypotension.

What makes a Stanford type A aortic dissection more dangerous than a [illegible]

How Do We Diagnose an Aortic Dissection? (LO #4)

In addition to the signs already described (acute chest or back pain, shock, neurologic deficits), typical physical exam, lab, and imaging findings are used to diagnose aortic dissection.

Physical Examination

If the dissection occludes one of the subclavian arteries, patients may have markedly unequal pulse amplitude or BPs between each arm (subclavian artery dissection). Less commonly, a difference between the arms and legs (in aortic dissections distal to the aortic arch) may be seen.

If the dissection dilates the aortic root and disrupts the aortic valve, the diastolic murmur of aortic insufficiency and a wide pulse pressure may be detected.

Rupture of the dissection into the pericardial space may cause signs of cardiac tamponade, such as hypotension, JVD, and pulsus paradoxus.

Imaging

The best test to confi [illegible] y.

CT Angiography. In stable patients, CT angiography is the preferred imaging modality (Figure 4).

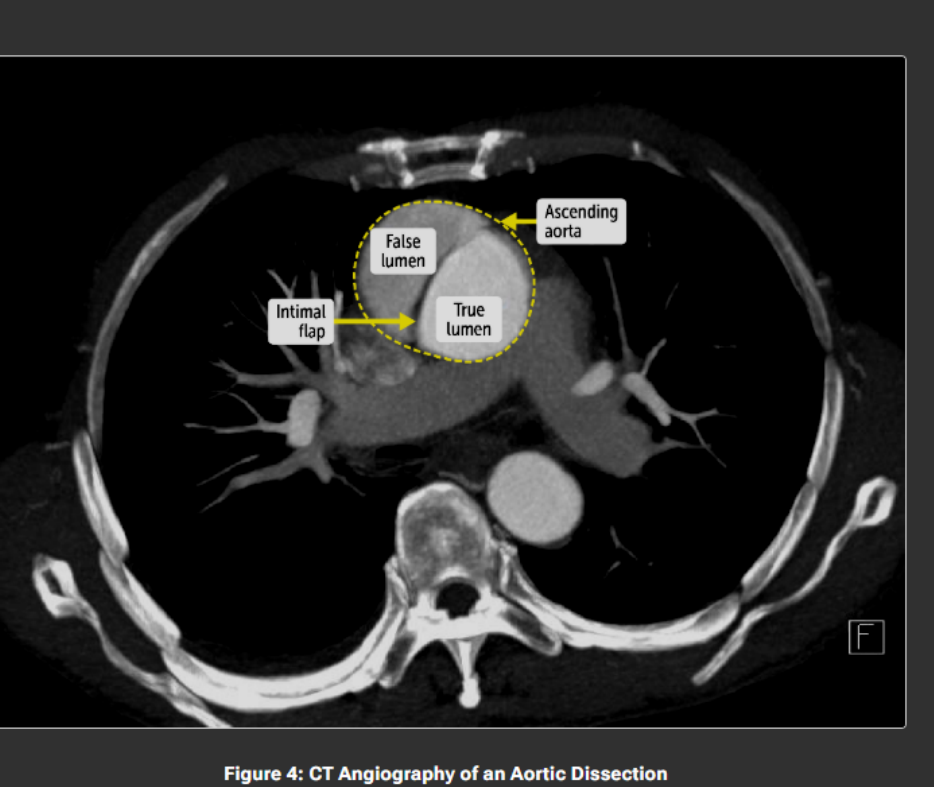


Figure 4: CT Angiography of an Aortic Dissection

The diagnosis is confirmed when an intimal flap is seen separating a false lumen from a true lumen on cross-sectional imaging. MR angiography is another option.

Transesophageal Ec [illegible] ents, this test is preferred because it can yield the diagnosis much faster and can be done at the bedside (Figure 5).

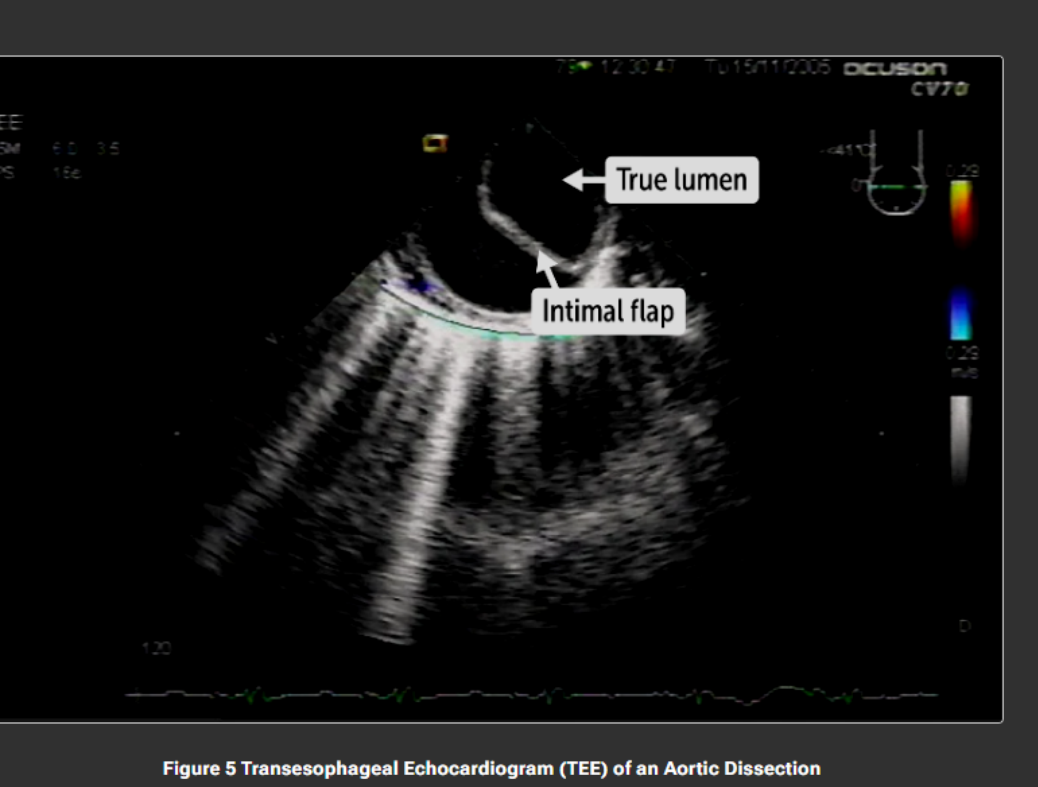


Figure 5 Transesophageal Echocardiogram (TEE) of an Aortic Dissection

CLINICAL CORRELATION

A disadvantage of TEE is that it requires esophageal intubation, which usually requires procedural sedation and may have untoward effects in hemodynamically unstable patients. T [illegible] ced operators to ensure accurate results.

Chest X-Ray. Chest x-ray has a low sensitivity for aortic dissection, but you may see **mediastinal widening**; however, this is absent in more than half of cases.

What is the imaging modality of choice for the diagnosis of aortic dissection in stable and unstable patients, respectively?

How Do We Manage Aortic Dissection? (LO #5)

An acute aortic dissection is a medical emergency, so patients generally require admission to the intensive care unit with intensive BP monitoring. Some patients go urgently to surgery.

Medications

Most patients with stable Stanford type B aortic dissections are treated medically to **decrease their BP**. This treatment reduces the shear forces on the intimal tear to minimize extension of the dissection. The best options are β-blockers and other antihypertensives. β-Blockers have the additional advantage of lowering the heart rate, which may improve prognosis. Target BP is less than 120/80 mm Hg, with pulse rates of less than 60 bpm.

Surgery

Endovascular or open surgical repair is used for all complicated dissections, such as those with organ ischemia, rapid degeneration, rupture, expansion, or refractory hypertension.

In addition, all Stanford type A aortic dissections merit immediate surgical treatment to repair the aorta.

What is the treatment for Stanford type A and type B aortic dissections, respectively?

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to KG, what diagnostic test would you advise? What treatment would you recommend?

Based on KG's history and examination, you suspect an aortic dissection. Because KG is stable, CT angiography is ordered, which reveals an intima media flap separating the false lumen from the true lumen, thus securing the diagnosis. The dissection is located in the descending thoracic aorta, with no involvement of the ascending aorta (Stanford type B). You know that this type of aortic dissection can be treated conservatively with medications, and you make sure KG has no contraindication to β-blockers. He is started on labetalol and observed carefully over the next several days.

Summary

- Aortic dissection is separation of the layers of the wall of the aorta.
- Aortic dissection is classified as Stanford type A if it affects the ascending aorta and Stanford type B if it does not.
- Aortic dissections usually present with sudden onset of severe pain that is described classically as "ripping" or "tearing" in character.
- Hypotension is more common in patients with Stanford type A aortic dissection and hypertension in those with Stanford type B aortic dissection.
- Aortic dissection [illegible] l degeneration of the tunica media.
- Consequences include rupture, blockage of blood flow (leading to stroke, myocardial infarction, or organ ischemia), aortic valve insufficiency, hemopericardium, and compression of adjacent structures.
- Physical exam might reveal unequal pulses in the two arms, hypotension, or hypertension.
- Diagnosis requires the use of CT angiography or transesophageal echocardiography.
- Stanford type A aortic dissections are managed surgically.
- Stanford type B aortic dissections are usually treated medically to lower blood pressure and pulse.

Review Questions

1. A 55-year-old male presents to the emergency department with severe, tearing chest pain radiating to his back. Imaging with contrast reveals an aortic dissection extending from the origin of the coronary arteries to the aorta's diaphragmatic entry. What is the diagnosis and treatment plan for this patient?

A. ○ Stanford type A aortic dissection, medical intervention with angiotensin-converting enzyme inhibitor
B. ○ Stanford type A aortic dissection, medical intervention with β-blocker
C. ● Stanford type A aortic dissection, surgical intervention
D. ○ Stanford type B aortic dissection, medical intervention with β-blocker
E. ○ Stanford type B aortic dissection, surgical intervention

Hide Explanation

The correct answer is Stanford type A aortic dissection, surgical intervention (C). Type A dissections require surgical intervention, not medical treatment with antihypertensive drugs, so Stanford type A aortic dissection, medical intervention with angiotensin-converting enzyme inhibitor (A) and Stanford type A aortic dissection, medical intervention with β-blocker (B) are incorrect. This patient's aortic dissection involves the ascending aorta and thus is Stanford type A, not type B, which makes Stanford type B aortic dissection, medical intervention with β-blocker (D) and Stanford type B aortic dissection, surgical intervention (E) incorrect.

2. Which of the following is the most important risk factor for aortic dissection?

A. ○ Female sex
B. ○ Heavy alcohol consumption
C. ● Hypertension
D. ○ Obesity
E. ○ Smoking

Hide Explanation

The correct answer is hypertension (C). Hypertension is the most important risk factor for aortic dissection because it is associated with the development of cystic medial degeneration. Female sex (A) is incorrect, as males are more likely to develop aortic dissection. Heavy alcohol consumption (B) has not been shown to be linked with aortic dissection. Obesity (D) is a significant risk factor for hypertension; however, it is not directly associated with aortic dissection. Smoking (E) may be associated with aortic dissection, but hypertension is a stronger risk factor.

3. A 72-year-old male is brought to the emergency department presenting with severe chest pain for the past 2 hours. His only medical condition is hypertension, but he has been nonadherent with medications. On examination, his systolic blood pressure is 55 mm Hg in the right arm and 75 mm Hg in the left arm. He has distant heart sounds and distended jugular veins. The patient dies soon after arrival. Which of the following is the most likely cause of death in this patient?

A. ● Hemorrhage into the pericardial space
B. ○ Hemorrhage into the retroperitoneal space
C. ○ Hemorrhage into the subarachnoid space
D. ○ Myocardial infarction
E. ○ Systemic vasodilation

Hide Explanation

The correct answer is hemorrhage into the pericardial space (A). This patient's presentation is most consistent with Stanford type A aortic dissection complicated by cardiac tamponade. This occurs when the dissection ruptures into the pericardiac space, causing hemopericardium, and sometimes tamponade with hypotension (from reduced cardiac output) and distant heart sounds. Hemorrhage into the retroperitoneal space (B) can be seen in a case of a ruptured abdominal aortic aneurysm, but this presentation is not consistent with the scenario described. Hemorrhage into the subarachnoid space (C) from a ruptured saccular aneurysm is often seen in patients with uncontrolled hypertension, but severe headache would be the primary symptom in these patients. Myocardial infarction (D) may be a complication of Stanford type A aortic dissections but would not cause distant heart sounds or unequal pulses. Systemic vasodilation (E) would be a cause of shock in systemic infections and would not cause the cardiac findings seen here.

4. For stable patients, what is the imaging modality of choice for the diagnosis of aortic dissection?

A. ○ Chest radiograph (Chest Xray)
B. ○ Nuclear stress test
C. ○ Transesophageal echocardiogram (TEE)
D. ○ Positron emission tomography
E. ● Compute [illegible]

Hide Explanation

The correct answer is CT angiogram (E). For stable patients, a CT angiogram is the preferred imaging modality for the diagnosis of an aortic dissection. A transesophageal echocardiogram (TEE) (C) has utility for less stable patients, but requires esophageal intubation. Chest Xray (A) has a low sensitivity for the diagnosis of aortic dissection. Nuclear stress test (B) and positron emission tomography (D) play no role in the diagnosis of aortic dissection.

How would you rate this Brick?

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Review Learning Objectives
Check off th [illegible] the concept
- Levy D, Goyal A, Grigorova Y, Farci F, Le JK. Aortic dissection. In: *StatPearls*. StatPearls Publishing; 2022.
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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the types of aortic dissection.~~
02 ☑ ~~Differentiate the clinical presentation of aortic dissection.~~
03 ☑ ~~Explain the pathophysiology and resulting consequences of aortic dissection.~~
04 ☑ ~~Interpret clinical findings to properly diagnose aortic dissection.~~
05 ☑ ~~Outline the management plan of aortic dissection.~~

Objective Confidence: 5/5